Four linear and unitary structural tables

Private L054 finite application

Theorem 1. *All 265 structural rows of the four current companion tables below have literal decomposition certificates for their actual numerical row groups, relative to the explicitly acquired named-model interfaces. The descriptions assert existence of the displayed decompositions, not uniqueness of a group with that extension label.*

Locator	Ambient description	Rows
`psl33`	$L_3(3)$	51
`psl33d2`	$L_3(3) : 2$	78
`psu33`	$U_3(3)$	36
`pgammau33`	$U_3(3) : 2$	100

Source and numerical inputs. The current PDFs are attributed to Thomas Connor and Dimitri Leemans. All eleven native pages were directly inspected. We acquire their accepted complete numerical inventories, exact source-row bijections and named ambient-model interfaces. Every imported subgroup is an actual literal permutation group: the saved arrays contain only `Group` expressions, with degrees $13, 104, 91, 91$ respectively. No printed group constructor is reconstructed inside a proper row. The complete earlier numerical classifications are not inferred afresh from the source counts.

Native products associate from the left unless parentheses say otherwise, as prescribed in the 2015 naming article. In `pgammau33`, row 3 is $3^{1+2} : (D_8 \cdot 2)$; its right factor is parenthesized. It is not $(3^{1+2} : D_8) \cdot 2$. Source 4^2 remains a direct power $C_4 \times C_4$, and 2^{1+4} and 3^{1+2} are the unsigned extraspecial atoms paid by the acquired registry. A dot, colon and cross retain their nonsplit, split and direct meanings. The inherited explicit Q_{12} dicyclic convention remains available, with no new nomenclature claim. Current PDFs are not assumed byte-identical to a historical edition.

Two named classical inputs. We explicitly retain the documented identities of the constructors `PSL(3,3)` and `PSU(3,3)`. GAP Reference Manual 50.2-12 and 50.2-14 specify the projective determinant-one linear and unitary groups, with the default permutation action on projective lines. These are construction interfaces, not isomorphism assertions deduced from order. Their original accepted complete executions are acquired: the exact drivers, clean successful receipts, all recorded input/output hashes and the resulting literal ambient generators are checked. Those generators are identical to the retained numerical top groups. They supply the concrete atoms $L_3(3)$ and $U_3(3)$. No new constructor, automorphism-group calculation or subgroup census is run. The extension rows are separately proved to have the indicated normal subgroup and complement by actual certificates.

Registry application. The application uses 69 acquired concrete structural models and the two named classical inputs. It adds 22 models. Every factor and occurrence map is an explicit bijection preserving identity and all right-generator edges. The acquired MC.3 word-induction criterion proves each map an isomorphism. Different models with the same structural name remain distinct registry entries until an actual map is supplied. Thus model-name matching alone pays no source row.

Proof of the finite application. The literal verifier closes every listed permutation group. New model references are acyclic. In each split model it checks the actual normal subgroup N, its already verified structural description, a subgroup K with the displayed complement description, $N \cap K = 1$, and $|N||K| = |H|$. The product is all of H. A noncommuting generator pair certifies the displayed nontrivial semidirect action. Normality and commutation tests on generators extend to the whole generated groups.

There are twenty new split and two new nonsplit models. The latter use the already proved composite-quotient lift criterion: for cyclic quotient of order six, fixed generator xN lifts to a complement exactly when some $a \in Nx$ has $a^6 = 1$; for quotient S_3 with fixed generators xN, yN, a complement exists exactly when some lifts satisfy $a^2 = b^3 = (ab)^2 = 1$. The checker first verifies the actual quotient relations and six cosets, then tests every required lift or lift pair. The $Q_8 \cdot 6$ model uses eight lift checks, and the $(D_8 : 2) \cdot S_3$ model uses 256 pairs. Their failure of the splitting relations proves nonsplitting, independently of any unsuccessful complement search.

Every source occurrence is transported to its exact embedded group under the acquired numerical row map. The source text, native syntax tree, order, subgroup index and generator arrays all agree. There are 307 checked transports and 146,010 right-generator edges. These checks prove the stated finite result, using only the explicit acquired interfaces.

Execution and serialization. One bounded candidate pass completed all 265 rows in 4.65 seconds. Normal-subgroup, complement and isomorphism software only proposed witnesses; the final proof uses their literal successful certificates. The replay does not repeat candidate discovery or numerical classification.

Two GAP output strings contain physical backslash–newline continuations between a comma and the next digit. The initial reader permitted only a continuation between two digits and stopped before building a certificate. That failed reader and the exact raw outputs are retained. The final reader permits only these two numeric positions, removes only the physical continuation, and parses the resulting JSON. All permutation entries, model references and equations are then checked normally. Malformed or nonnumeric continuations are rejected. No scientific output was changed.

Controls and scope. Damaged certificates test missing/duplicate rows, incorrect identity or right-generator edges, changed subgroup generators and row maps, source exponents, dots and parentheses, wrong complements, cyclic factor references, and both new quotient witnesses. Changed named-classical generators fail the exact acquired-input check. The actual constructor-acquisition routine also rejects incomplete execution, wrong constructor, bad return code and changed-input receipts. Native-expression controls distinguish the parenthesized right factor and reject unmatched or empty parentheses. The existing independent prime-index, direct-product, dicyclic and composite-quotient small controls remain executable. Ordinary and optimized Python must produce identical complete reports.

The result is 265 current structural row joins. Numerical completeness, named constructors and earlier model proofs are retained inputs; no independent reproof of those software interfaces is claimed. There is no new ambient census, unique-name assertion, historical-byte claim, whole-original disposition, question, formalization or measurement credit. The finite application and source-sensitive implementation are authored here under root's registry-reuse task and await distinct review.